- ***Attrition Intelligence Dashboard***

Business Scenario:

An HR department wants to overhaul its reporting by creating a full-journey dashboard from hiring to resignation. The aim is to improve talent retention and manager effectiveness.

Power BI Focus:

- Hierarchical drilldowns (company → department → employee)
- Predictive metrics using DAX
- Multi-page report storytelling

STAR technique

SITUATION

In our ongoing efforts to understand employee attrition and identify trends, we created a comprehensive table containing all the details of every employee in the company. However, mere raw numbers couldn't explain the concerning factors behind people leaving their jobs.

TASK

Therefore, my task was to uncover insights by comparisons across different parameters, and suggest data–driven solutions, so that the company can make informed decisions to enhance fairness, motivation, and retention.

ACTION

1. Energize onboarding & engagement for new hires (technical & sales roles).
2. Compensation/promotion/expectations (high-demanding roles like Sales Rep and Lab Tech).
3. Flexible work culture (remote/hybrid).
4. Clear overtime expectations - Optimize workflows.

5. Managerial training on work-life sensitivity.
6. Blind performance reviews, standardize hike criteria across all genders. Regular auditing payments by job role and department.
7. Engage newcomers with meaningful tasks to build ownership.
8. Friendly environment - team interactions in tea breaks or small team lunches.
9. Clarify roles and team goals early. Reward achievements as "Employee of the Month/Quarter" via social media shout-outs.
10. Employee pulse surveys, eNPS, and exit interviews to identify root cause.

Objective:

Questions with their answers:

1. Which combinations of Job Role, Department, and Job Level show the highest voluntary attrition rates?

Department – Sales, Job role – Sales Representative, Job level – 1 and 2, exhibits the highest attrition rate (40% of the total 237 attrition population). There is no information available about the type of resignation (voluntary or involuntary) in the dataset. Therefore, can't comment on that.

2. How do these rates differ by Business Travel frequency and Overtime status?

Total Employees – 1470 (100%), Active Employees – 1233 (84%), Attrition Population – 237 (16%).

Out of 237 people who left their jobs,
66% - rarely travel (**35.5%** - overtime)
29% - frequently travel (**15%** - overtime)
5% - have desk jobs (**3%** - overtime)

~ 53% out of 237 reported doing **overtime,** which points towards being one of the major reasons for attrition.

3. Using the available features (Age, Distance from Home, Job Satisfaction, Environment Satisfaction, etc.), which top five variables most strongly explain whether an employee leaves?

Attrition is predominantly influenced by

1. **Poor work-life balance,**
2. **Low job involvement**
3. **Low Job satisfaction**
4. **Low Environmental satisfaction**
5. **Age Group (Especially in the 30 – 45 age group)**

4.

I. How does the risk profile change across Age brackets (e.g., <30, 30-45, >45)?

Among younger or mid-career employees **(30 – 45 age group)** in entry-level roles, attrition is heavily concentrated (**47%**) at junior levels **(job level 1 & 2)**, possibly due to limited career growth, compensation, or poor onboarding. The risk decreases to **38%** within the younger mass, i.e., **age 18 – 30,** and further decreases **to ~15%** within the elderly age group, i.e., the **46 – 60 age** bracket. This resonates with the **KPI** that says the **average attrition age** is **34**.

II. How does the risk profile change across Education Levels?

High Representation = High Exposure to Attrition

- **Education Level 3 has the highest number of employees (572 / 38.9%)**, and also the **highest attrition count (99 / 41.8%)**. Naturally, a larger workforce segment increases the **base for potential exits**, skewing the attrition numbers higher.
- **Education Level 2 & 1**: fairly balanced employee and attrition shares, with **slightly higher attrition headcount in Level 1.**
- **Level 4** shows **better retention**, with a lower attrition rate (24.5%) compared to its workforce share (27.1%).
- **Level 5**, though the smallest group, shows the **strongest retention,** indicating **better compensation** at senior levels.

Call to Action

1. Investigate the cause of **high attrition among Education Level 3** — (role mismatch, lack of growth, or compensation). Support mid-

educated employees with career pathing, incentives, and engagement programs.

2. **Leverage Level 4's stability** as a benchmark for retention practices.
3. Monitor **Level 1 and 2** closely to prevent future attrition surges.
4. Engage and retain **Level 5** through continued challenge and leadership opportunities.

5. Is higher Job Involvement or Relationship Satisfaction associated with better Performance Ratings?

No. Because the data is primarily skewed toward **Performance Rating 3** (on a scale of **1 – 4** in **Performance Rating**, most employees have 3 and only a few have 4), and the **Relationship Satisfaction** does not appear to significantly influence the distribution of performance ratings in this sample.

6. At what point do increases in Training Times Last Year stop yielding higher performance?

A weak **negative relationship** between training sessions and performance rating. Despite slight fluctuations (**3.13 to 3.19**), average ratings remain stable. The DAX calculated **Pearson's correlation coefficient** is about -**0.02**, indicating a weak negative relationship: as the number of training sessions increases, the average performance rating tends to decrease a little, but the effect is not strong.

Negative Correlation Meaning:
More training does not always lead to higher performance in this dataset.

This could suggest:

- Training programs are ineffective or not aligned with job requirements.
- High performers may not need to attend as many training sessions (e.g., they are already skilled).

- Other factors (e.g., workload, motivation) may influence both training attendance and performance.

7. For similar Job Roles and Job Levels, do Monthly Income and Percent Salary Hike differ significantly by Gender or Marital Status?

<u>Key Observations:</u>

By Gender:

Monthly Income: Males earn slightly more in high-paying roles (Manager, Director), but the differences are small. Females earn slightly more in lower-paying roles (e.g., Research Scientist, Sales Representative, Manufacturing Director, Human Resources).

Salary Hike%: Males receive notably higher hikes in most roles. The Gap can span from 2 to 3% lower in Females than their Male counterparts, especially in Manager roles (HR and R&D).

Overall, **Gender does impact** pay and hikes subtly, favouring males in many roles, though differences are relatively minor.

By Marital Status:

Single employees, particularly in Managerial and Technical roles, receive better hikes, indicating higher reward flexibility.

Married employees get slightly higher average base salaries, especially in Manager roles (52.9K), but less so in entry-level roles. seem to get stable but moderate hikes, suggesting seniority might be rewarded more in base pay than increments. Managerial roles see a lower hike % (max ~15%) than Singles.

For **Divorced employees**, base pay is also high (e.g., 50 K+ for Managers, 16 K+ for Directors), on par with or slightly lower than married. Salary Hike% is generally lowest, especially in the Manager role (~13%). Hikes are less aggressive across all roles. Divorced employees might be valued more for experience (base pay) but less prioritized during salary reviews and hikes.

8. How does Total Working Years or Years Since Last Promotion influence current pay?

- Influence of Total Working Years on Monthly Income**: Strong Positive Correlation**
  Experience matters — employees with longer total working years earn substantially more. Likely due to accumulated skills, promotions, and tenure benefits.

- Influence of Years Since Last Promotion on Monthly Income: **Positive Correlation**
  Even without recent promotions, employees still receive salary increases, suggesting alternative compensation mechanisms (e.g., merit hikes, role expansions).

9. Does a lower Work-Life Balance score correlate with greater Overtime frequency and higher attrition?

1. Employees who work overtime are slightly more likely to leave the organization.
2. Although both categories show notable attrition, working overtime is associated with higher attrition.
3. **Most attrition** occurs among employees with a **Work-Life Balance rating of 3,** which may represent the most common rating in the dataset.
4. **Very few attritions** come from those with **very low (1)** or **very high (4)** work-life balance scores. This implies that work-life balance alone isn't the dominant cause of attrition, but other factors likely intersect with it.

10. Which departments manage to keep both high Work-Life Balance and low attrition simultaneously?

Research Director - Shows above average Work-Life Balance (**2.86** on a scale of 1 – 4, where **the average Work-Life Balance** is **2.76**) and has the lowest attrition (2 out of 237), simultaneously.

11. How long, on average, do employees stay in their Current Role before their next promotion, and how does that vary by Job Level?

1. Employees stay on average about **6-10** years in their role before their next promotion.
2. Higher job levels (especially Levels 4 and 5) generally have longer average durations, indicating slower progression at higher levels due to the last step of the Promotion Ladder.
3. The data across all images is consistent, showing a clear trend of increasing tenure with higher job levels.

12. What share of high-tenure employees (≥10 Years at Company) still receive regular promotions or salary hikes?

35% share of high-tenure employees (≥10 Years at Company) still receive regular promotions or salary hikes.

The DAX used below finds the percentage of employees who have at least 10 years in the job and have taken at least 1 promotion with a positive salary percentage hike.

DAX:

```
Share of High Tenure Employees =
FORMAT(
   DIVIDE(
      CALCULATE(
COUNT(AttritionParameters[EmployeeNumber]),NumericalFactors[TotalWorkingYears] >= 10, NumericalFactors[YearsSinceLastPromotion] > 0, NumericalFactors[PercentSalaryHike] > 0),
      COUNT(AttritionParameters[EmployeeNumber]),0),
"#,#%")
```

Deliverables:

- Attrition heatmaps and forecasting indicators ✅
- Correlation matrix between job roles, satisfaction, and turnover ✅
- HR KPI report with filters by tenure and department ✅

RESULT

By leveraging this dashboard, HR and leadership can now pinpoint where reward mechanisms may need recalibration, ensuring motivation, retention, and fairness across the board. This is not just about numbers, but about people and potential.

LIST OF CHARTS USED IN THE DASHBOARD

1. Page 1
   (Q 1 & 2) – Table with Conditional Column, Pie Chart, KPI Cards, Slicers
2. Page 2
   (Q 3 & 4) – Funnel Charts, Bar Chart
3. Page 3
   (Q 5 & 10) – Table with Conditional Column
4. Page 4
   (Q 6) – Line chart

5. Page 5
   (Q 7) – Column Charts, Slicers
6. Page 6
   (Q 8.1) – Waterfall Chart, Slicer
7. Page 7
   (Q 8.2) – Waterfall Chart, Slicer
8. Page 8
   (Q 9) – Ribbon Charts, Bubble Chart
9. Page 10
   (Q 11 & 12) – Area Chart, Card, Slicer
10. Page 11
   (Q 5 – Details) – Table with Conditional Column
11. Page 12
   Extra Information (May SKIP) – Column Charts

LIST OF COLUMNS IN THE DATASET

Age: Age of employee.
Attrition: attrition status.
Travel business: Working conditions in the form of frequent or not frequent business travel.
Daily rate: the payment a person receives for each day of work performed.
Department: A unit of work.
Distance from home: The distance from an employee's home to the office.
Education: Level of education.
Education field: Education major.
Employee count: Totals of employees.
Environment satisfaction: Level of satisfaction with the work environment.
Gender: gender.
Hourly rate: Hourly wage.
Job involvement: The extent to which an employee feels attached and committed to his/her job and the organization he/she works for.
Job level: The career level of a job.
Job role: The name of the job position within the company.
Job satisfaction: Satisfaction in the field of work.
Marital status: Relationship status whether married, divorced or single.
Monthly income: Monthly income.
Over time: Overtime duration.

Percent salary hike: percentage of wage increase.
Performance rating: performance on the job.
Relationship satisfaction: satisfaction in relationship.
Standard hours: Normal working hours.
Stock option level: Stock distribution based on position level.
Total working years: Total working years.
Training time last year: Duration of training in the previous year.
Work life balance: Level of satisfaction in life.
Years at company: Number of years working for this company.
Years in current role: Number of years working in current role.
Years since last promotion: Number of years since last promotion.
Years with curr manager: Number of years working with current manager.